Page 1: Operating System Fundamentals

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Page 2: Processes and Process Management

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Page 3: CPU Scheduling

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Page 4: Threads and Concurrency

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Page 5: Process Synchronization

Synchronization coordinates concurrent processes and threads. The critical-section problem requires mutual exclusion, progress, and bounded waiting. Semaphores can coordinate resource access, while mutexes provide exclusive ownership of protected regions. Producer-consumer, readers-writers, and dining-philosophers problems illustrate synchronization issues. Poor synchronization can cause races, deadlocks, starvation, or unnecessary blocking. Blocking mechanisms allow waiting processes to sleep rather than wasting processor time through busy waiting. Operating-system design involves abstraction, resource management, protection, and controlled sharing. Different components cooperate through well-defined interfaces. A scheduler selects work, a memory manager controls address spaces, a file system organizes persistent data, and device managers coordinate hardware. The kernel must preserve system invariants while responding to concurrent requests. Measurements such as latency, throughput, utilization, and fault rates help evaluate behavior under different workloads. An important learning strategy is to connect definitions with concrete execution steps. A process moves through states because events occur. A page fault causes a

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Page 6: Deadlocks

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Page 7: Memory Management

Memory management allocates and protects main memory for processes. Contiguous allocation can produce internal or external fragmentation. Paging divides logical memory into pages and physical memory into frames. Page tables map virtual pages to physical frames, and a translation lookaside buffer can accelerate translation. Segmentation represents logical regions such as code, data, and stack. Memory protection prevents processes from accessing unauthorized regions. Operating-system design involves abstraction, resource management, protection, and controlled sharing. Different components cooperate through well-defined interfaces. A scheduler selects work, a memory manager controls address spaces, a file system organizes persistent data, and device managers coordinate hardware. The kernel must preserve system invariants while responding to concurrent requests. Measurements such as latency, throughput, utilization, and fault rates help evaluate behavior under different workloads. An important learning strategy is to connect definitions with concrete execution steps. A process moves through states because events occur. A page fault causes a

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Page 8: Virtual Memory

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Page 9: Page Replacement

Page replacement selects a page to remove when a new page must be loaded and no free frame is available. FIFO removes the oldest page and is simple but can exhibit Belady's anomaly. Optimal replacement removes the page whose next reference is farthest away and gives the theoretical minimum for a known reference sequence. LRU removes the least recently used page. Second Chance and Clock algorithms approximate recent-use behavior with lower implementation cost. Operating-system design involves abstraction, resource management, protection, and controlled sharing. Different components cooperate through well-defined interfaces. A scheduler selects work, a memory manager controls address spaces, a file system organizes persistent data, and device managers coordinate hardware. The kernel must preserve system invariants while responding to concurrent requests. Measurements such as latency, throughput, utilization, and fault rates help evaluate behavior under different workloads. An important learning strategy is to connect definitions with concrete execution steps. A process

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Page 10: File Systems

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Page 11: Operating System Fundamentals

An operating system manages computer hardware and provides services to application programs. It controls processors, memory, storage, input and output devices, files, security mechanisms, and communication facilities. The kernel performs privileged operations and provides controlled interfaces to applications. System calls allow programs to request services without directly manipulating hardware. Operating systems create abstractions such as processes, files, virtual memory, sockets, and devices so that programmers can work with resources through consistent interfaces. Operating-system design involves abstraction, resource management, protection, and controlled sharing. Different components cooperate through well-defined interfaces. A scheduler selects work, a memory manager controls address spaces, a file system organizes persistent data, and device managers coordinate hardware. The kernel must preserve system invariants while responding to concurrent requests. Measurements such as latency, throughput, utilization, and fault rates help evaluate behavior under different workloads. An important learning strategy is to connect definitions with concrete execution steps. A process moves

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Page 12: Processes and Process Management

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Page 13: CPU Scheduling

CPU scheduling determines which ready process receives the processor. Common measurements include CPU utilization, throughput, turnaround time, waiting time, response time, and fairness. First-Come First-Served follows arrival order. Shortest Job First selects the shortest expected CPU burst. Priority scheduling uses priority values. Round Robin gives processes time quanta and supports responsive multitasking. Multilevel and feedback queue algorithms divide workloads into different scheduling classes. Operating-system design involves abstraction, resource management, protection, and controlled sharing. Different components cooperate through well-defined interfaces. A scheduler selects work, a memory manager controls address spaces, a file system organizes persistent data, and device managers coordinate hardware. The kernel must preserve system invariants while responding to concurrent requests. Measurements such as latency, throughput, utilization, and fault rates help evaluate behavior under different workloads. An important learning strategy is to connect definitions with concrete execution steps. A process moves through states because events occur. A page fault causes

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Page 14: Threads and Concurrency

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Page 15: Process Synchronization

Synchronization coordinates concurrent processes and threads. The critical-section problem requires mutual exclusion, progress, and bounded waiting. Semaphores can coordinate resource access, while mutexes provide exclusive ownership of protected regions. Producer-consumer, readers-writers, and dining-philosophers problems illustrate synchronization issues. Poor synchronization can cause races, deadlocks, starvation, or unnecessary blocking. Blocking mechanisms allow waiting processes to sleep rather than wasting processor time through busy waiting. Operating-system design involves abstraction, resource management, protection, and controlled sharing. Different components cooperate through well-defined interfaces. A scheduler selects work, a memory manager controls address spaces, a file system organizes persistent data, and device managers coordinate hardware. The kernel must preserve system invariants while responding to concurrent requests. Measurements such as latency, throughput, utilization, and fault rates help evaluate behavior under different workloads. An important learning strategy is to connect definitions with concrete execution steps. A process moves through states because events occur. A page fault causes a

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Page 16: Deadlocks

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Page 17: Memory Management

Memory management allocates and protects main memory for processes. Contiguous allocation can produce internal or external fragmentation. Paging divides logical memory into pages and physical memory into frames. Page tables map virtual pages to physical frames, and a translation lookaside buffer can accelerate translation. Segmentation represents logical regions such as code, data, and stack. Memory protection prevents processes from accessing unauthorized regions. Operating-system design involves abstraction, resource management, protection, and controlled sharing. Different components cooperate through well-defined interfaces. A scheduler selects work, a memory manager controls address spaces, a file system organizes persistent data, and device managers coordinate hardware. The kernel must preserve system invariants while responding to concurrent requests. Measurements such as latency, throughput, utilization, and fault rates help evaluate behavior under different workloads. An important learning strategy is to connect definitions with concrete execution steps. A process moves through states because events occur. A page fault causes a

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Page 18: Virtual Memory

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Page 19: Page Replacement

Page replacement selects a page to remove when a new page must be loaded and no free frame is available. FIFO removes the oldest page and is simple but can exhibit Belady's anomaly. Optimal replacement removes the page whose next reference is farthest away and gives the theoretical minimum for a known reference sequence. LRU removes the least recently used page. Second Chance and Clock algorithms approximate recent-use behavior with lower implementation cost. Operating-system design involves abstraction, resource management, protection, and controlled sharing. Different components cooperate through well-defined interfaces. A scheduler selects work, a memory manager controls address spaces, a file system organizes persistent data, and device managers coordinate hardware. The kernel must preserve system invariants while responding to concurrent requests. Measurements such as latency, throughput, utilization, and fault rates help evaluate behavior under different workloads. An important learning strategy is to connect definitions with concrete execution steps. A process

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Page 20: File Systems

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Page 21: Operating System Fundamentals

An operating system manages computer hardware and provides services to application programs. It controls processors, memory, storage, input and output devices, files, security mechanisms, and communication facilities. The kernel performs privileged operations and provides controlled interfaces to applications. System calls allow programs to request services without directly manipulating hardware. Operating systems create abstractions such as processes, files, virtual memory, sockets, and devices so that programmers can work with resources through consistent interfaces. Operating-system design involves abstraction, resource management, protection, and controlled sharing. Different components cooperate through well-defined interfaces. A scheduler selects work, a memory manager controls address spaces, a file system organizes persistent data, and device managers coordinate hardware. The kernel must preserve system invariants while responding to concurrent requests. Measurements such as latency, throughput, utilization, and fault rates help evaluate behavior under different workloads. An important learning strategy is to connect definitions with concrete execution steps. A process moves

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Page 22: Processes and Process Management

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Page 23: CPU Scheduling

CPU scheduling determines which ready process receives the processor. Common measurements include CPU utilization, throughput, turnaround time, waiting time, response time, and fairness. First-Come First-Served follows arrival order. Shortest Job First selects the shortest expected CPU burst. Priority scheduling uses priority values. Round Robin gives processes time quanta and supports responsive multitasking. Multilevel and feedback queue algorithms divide workloads into different scheduling classes. Operating-system design involves abstraction, resource management, protection, and controlled sharing. Different components cooperate through well-defined interfaces. A scheduler selects work, a memory manager controls address spaces, a file system organizes persistent data, and device managers coordinate hardware. The kernel must preserve system invariants while responding to concurrent requests. Measurements such as latency, throughput, utilization, and fault rates help evaluate behavior under different workloads. An important learning strategy is to connect definitions with concrete execution steps. A process moves through states because events occur. A page fault causes

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Page 24: Threads and Concurrency

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Page 25: Process Synchronization

Synchronization coordinates concurrent processes and threads. The critical-section problem requires mutual exclusion, progress, and bounded waiting. Semaphores can coordinate resource access, while mutexes provide exclusive ownership of protected regions. Producer-consumer, readers-writers, and dining-philosophers problems illustrate synchronization issues. Poor synchronization can cause races, deadlocks, starvation, or unnecessary blocking. Blocking mechanisms allow waiting processes to sleep rather than wasting processor time through busy waiting. Operating-system design involves abstraction, resource management, protection, and controlled sharing. Different components cooperate through well-defined interfaces. A scheduler selects work, a memory manager controls address spaces, a file system organizes persistent data, and device managers coordinate hardware. The kernel must preserve system invariants while responding to concurrent requests. Measurements such as latency, throughput, utilization, and fault rates help evaluate behavior under different workloads. An important learning strategy is to connect definitions with concrete execution steps. A process moves through states because events occur. A page fault causes a

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Page 26: Deadlocks

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Page 27: Memory Management

Memory management allocates and protects main memory for processes. Contiguous allocation can produce internal or external fragmentation. Paging divides logical memory into pages and physical memory into frames. Page tables map virtual pages to physical frames, and a translation lookaside buffer can accelerate translation. Segmentation represents logical regions such as code, data, and stack. Memory protection prevents processes from accessing unauthorized regions. Operating-system design involves abstraction, resource management, protection, and controlled sharing. Different components cooperate through well-defined interfaces. A scheduler selects work, a memory manager controls address spaces, a file system organizes persistent data, and device managers coordinate hardware. The kernel must preserve system invariants while responding to concurrent requests. Measurements such as latency, throughput, utilization, and fault rates help evaluate behavior under different workloads. An important learning strategy is to connect definitions with concrete execution steps. A process moves through states because events occur. A page fault causes a

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Page 28: Virtual Memory

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Page 29: Page Replacement

Page replacement selects a page to remove when a new page must be loaded and no free frame is available. FIFO removes the oldest page and is simple but can exhibit Belady's anomaly. Optimal replacement removes the page whose next reference is farthest away and gives the theoretical minimum for a known reference sequence. LRU removes the least recently used page. Second Chance and Clock algorithms approximate recent-use behavior with lower implementation cost. Operating-system design involves abstraction, resource management, protection, and controlled sharing. Different components cooperate through well-defined interfaces. A scheduler selects work, a memory manager controls address spaces, a file system organizes persistent data, and device managers coordinate hardware. The kernel must preserve system invariants while responding to concurrent requests. Measurements such as latency, throughput, utilization, and fault rates help evaluate behavior under different workloads. An important learning strategy is to connect definitions with concrete execution steps. A process

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Page 30: File Systems

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